

KAMEX

Carrier Gateway Operations Console

Detailed UI/UX Functional Specification & Use Case Flows

Purpose: Guide a thorough redesign of the operational interface for a Kannel-derived SMS gateway.

Scope: Carrier connectivity, SMSC/SMPP operations, traffic, queues, DLRs, routing, diagnostics, service monitoring and audit.

Explicitly out of scope: CPaaS customer provisioning, commercial billing, plans, customer API credentials and customer portal functions.

1. Product Definition

Kamex UI is the operational control plane for engineers and NOC teams running the Kamex messaging gateway. The redesign should hide Kannel configuration-file concepts from normal operators and instead present domain concepts that match telecom operations: Carrier → SMSC → SMPP Session → Route → Queue → Message → Delivery Receipt.

The interface must optimize for rapid situational awareness, safe operational intervention, fast fault isolation, and traceability. Kannel/Kamex internals may remain available to backend services and privileged engineering diagnostics, but should not define the primary user experience.

1.1 Design principles

- Operations first: every major screen answers health, impact, cause and next action.
- Progressive disclosure: show status and actionable summaries first; expose protocol detail on drill-down.
- No routine file editing: configuration files, raw Kannel groups and directives are not the primary UI abstraction.
- Safe control: disruptive actions require permissions, impact warnings, reason capture and audit logging.
- Correlation by default: connect alerts, carrier events, SMPP errors, queue changes, message traces and logs.
- Real-time but calm: live data should update without visually destabilizing tables or charts.
- Privacy aware: mask MSISDN/message content by default; unmasking is privileged and audited.
- NOC friendly: usable on desktop workstations and large wall displays, with dark-mode readiness.

1.2 Personas and roles

Persona	Primary goals	Typical privileges
NOC Operator	Detect outages, monitor queues/TPS, acknowledge alerts, perform approved recovery actions	Read operational data; reconnect/resume; acknowledge alerts
Messaging Engineer	Troubleshoot SMPP, routing, DLR and carrier behavior	Deep diagnostics; tests; route/failover controls
Gateway Administrator	Manage operational connectivity objects and service controls	Create/update SMSC operational metadata; service actions; thresholds
Support Engineer	Trace individual messages and explain failures	Message trace, DLR/error lookup, masked logs
Auditor / Read-only	Review availability, incidents and actions	Read-only dashboards, events and audit trail

1.3 Scope boundary with CPaaS console

Kamex owns	CPaaS console owns
Carrier/SMSC connectivity	Client/customer onboarding
SMPP sessions and health	Subscriptions, packages and billing
Network/carrier routing and failover	Customer-facing routing products/policies
Gateway queues and throughput	Customer quotas and commercial limits
DLR operational quality	Customer reporting/portal analytics
Message diagnostics	Customer API credentials
Gateway/service monitoring	Commercial sender onboarding
Operational alerts and audit	Invoices, payments and account lifecycle

2. Information Architecture

Recommended primary navigation:

Section	Screens	Purpose
Overview	Dashboard; Alerts	Global health and active operational risk
Connectivity	Carriers; SMSCs; SMPP Sessions	Carrier-facing connectivity and session operations
Traffic	Live Traffic; Queues; DLR Performance	Real-time message flow and congestion
Routing	Carrier Routes; Failover; Route Simulator	Network route behavior and continuity

Diagnostics	Message Trace; SMPP Errors; Events; Logs; Test Tools	Fault isolation and investigation
System	Services; Nodes; Performance; Audit Trail	Gateway platform health and accountability

2.1 Global UI shell

- Persistent environment indicator (e.g., Production / DR / Staging) with strong visual distinction.
- Global time-range control for analytical screens; display gateway timezone and allow user-local rendering.
- Global search for carrier, SMSC, session, message ID and MSISDN where permitted.
- Alert bell with severity counts and direct incident navigation.
- Connection/live-data indicator showing whether telemetry is current, delayed or disconnected.
- User/role menu and audit-aware privileged-mode indicator.
- Breadcrumbs that preserve hierarchy, e.g., Carriers / MTN Uganda / MTN-P1 / TRX-02.

3. Dashboard Specification

3.1 Purpose

Provide a 10-second assessment of whether the gateway and carrier estate are healthy, where impact exists, and which issue needs attention first.

3.2 Recommended layout

Zone	Components	Interaction
Health strip	Gateway health; SMSCs up/total; current MT TPS; MO TPS; queue; DLR success; active critical alerts	Click metric to open filtered detail
Carrier connectivity	Carrier cards/table with health, sessions, TPS, queue, DLR	Click carrier for carrier overview
Traffic trend	MT/MO/DLR time series; current vs normal band where available	Brush/zoom; filter carrier/SMSC
Queue pressure	Top queues by depth, age and growth	Open queue; show estimated drain time
Carrier quality	Delivery rate, P95 DLR latency, reject/throttle rate	Rank worst performers first
Active incidents	Critical/warning alerts with duration, impact and acknowledgement	Open alert detail / acknowledge
System health	Core services/nodes and telemetry freshness	Drill to System

3.3 Health semantics

- Healthy: service/SMSC is available and key performance indicators remain inside defined thresholds.
- Degraded: available but experiencing elevated errors, latency, queue growth, session loss or capacity pressure.
- Critical: unavailable or materially unable to process expected traffic.
- Unknown: telemetry stale or insufficient; never represent unknown as healthy.

4. Carrier & SMSC Management

4.1 Carrier object

Carrier is an operational aggregation object representing a mobile network/operator and its associated SMSCs and sessions.

Carrier overview element	Required data
Identity	Carrier name, country/market, network code/metadata, operational status
Connectivity	SMSC count, healthy/failed sessions, last connectivity event
Capacity	Configured/known capacity, current TPS, utilization, throttling

Quality	Delivery %, rejection %, DLR median/P95, no-DLR rate
Traffic	MT, MO and DLR volumes/TPS
Incidents	Open alerts and recent carrier events

4.2 SMSC list

Column	Behavior
SMSC	Name/ID; click to detail
Carrier	Parent carrier
Protocol	SMPP or other supported connector
State	Connected / degraded / disconnected / suspended
Sessions	Healthy / total
Current TPS	Outbound and optional inbound
Capacity	Known/configured ceiling and utilization
Queue	Depth plus oldest age
DLR	Delivery rate and latency indicator
Last event	Disconnect/rebind/throttle timestamp
Actions	Context menu based on role and current state

UC-CON-01 — Investigate an unhealthy carrier

Field	Specification
Primary actor	NOC Operator / Messaging Engineer
Trigger	Carrier card turns degraded/critical or an alert links to the carrier.
Preconditions	Telemetry is available and user has read permission.
Postconditions	Carrier remains monitored; any action or acknowledgement is recorded.

Main flow

1. User opens the carrier from Dashboard or Connectivity.
2. System shows aggregate health and highlights unhealthy SMSCs/sessions.
3. User reviews the synchronized traffic, queue, DLR and error trends around the incident time.
4. User opens the affected SMSC and inspects session state and recent events.
5. User follows a correlated event/error to diagnostics or logs when deeper evidence is required.
6. User either performs an authorized recovery action or records/acknowledges the incident.

Exceptions / alternate flows

- If telemetry is stale, show Unknown state and telemetry warning rather than inferred health.
- If multiple SMSCs are degraded, rank them by traffic impact and queue growth.

UI requirements

- Sticky incident context/time window while drilling down.
- Show 'what changed' summary: session loss, queue increase, delivery drop, throttling spike.
- Provide related alerts/events in the same view.

UC-SMSC-01 — Reconnect an SMSC

Field	Specification
Primary actor	Authorized NOC Operator / Gateway Administrator
Trigger	Operator determines an SMSC requires reconnection.
Preconditions	SMSC exists; user has reconnect permission; gateway control channel is healthy.
Postconditions	SMSC is either connected/degraded or remains failed with a recorded result.

Main flow

7. User opens SMSC detail and selects Reconnect.
8. System displays current state, affected sessions, queue depth and estimated operational impact.
9. User enters/selects a reason and confirms.
10. System sends the controlled reconnect command and immediately changes action state to 'In progress'.

11. UI streams resulting events: disconnect, bind attempt, bind result, health checks.
12. System reports success or failure and refreshes traffic/session health.
13. Audit trail records actor, reason, timestamps, prior state and result.

Exceptions / alternate flows

- If a reconnect is already in progress, disable duplicate action.
- If the control service is unavailable, fail safely without changing displayed state.
- If bind fails, present the protocol response and suggested diagnostic links.

UI requirements

- Never use optimistic 'Connected' status until bind/health telemetry confirms it.
- Confirmation dialog must show impact, not just 'Are you sure?'.
- Provide one-click navigation to related events/logs after failure.

UC-SMSC-02 — Suspend and resume traffic

Field	Specification
Primary actor	Messaging Engineer / Gateway Administrator
Trigger	Planned maintenance, carrier request, abnormal behavior or incident response.
Preconditions	User has traffic-control permission.
Postconditions	Traffic is intentionally suspended or resumed with full audit history.

Main flow

14. User chooses Suspend Traffic on the SMSC.
15. System explains whether suspension stops new submissions only or affects sessions, according to backend capability.
16. User provides reason and confirms.
17. SMSC receives a prominent Suspended badge; queue and incoming rate remain visible.
18. When safe, user selects Resume Traffic.
19. System confirms state transition and monitors queue drain and error rate.

Exceptions / alternate flows

- If queue growth becomes critical while suspended, create/escalate an alert.

UI requirements

- Clearly distinguish operator-suspended from carrier-disconnected states.

5. SMPP Session Monitoring

Session field	Display/behavior
Bind state	OPEN, BOUND_TX/RX/TRX, UNBOUND, CLOSED; map to human-readable status
Endpoint	Remote IP/host and port; mask credentials completely
Uptime	Current bind/session uptime
Last activity	Last protocol activity timestamp
Enquire link	Last request/response, round-trip time, missed responses
Counters	submit_sm, submit_sm_resp, deliver_sm, deliver_sm_resp, generic_nack
Latency	Protocol response latency median/P95 where available
Reconnects	Count over selected time range
Timeouts	Count/rate and latest occurrence
Errors	Top SMPP command status codes

UC-SMPP-01 — Diagnose repeated SMPP disconnects

Field	Specification
Primary actor	Messaging Engineer
Trigger	Alert indicates session flapping or operator notices frequent reconnects.

Preconditions	Session telemetry and event history exist.
Postconditions	Likely fault domain is identified and evidence is available for remediation/escalation.

Main flow

20. User opens affected SMPP session.
21. System displays bind timeline and reconnect frequency.
22. User aligns disconnects with enquire_link timeouts, socket errors, SMPP responses and traffic level.
23. User opens a disconnect event to see protocol/network context.
24. User compares sibling sessions on the same SMSC to determine whether failure is session-specific or carrier-wide.
25. User exports/copies a concise incident diagnostic summary for carrier escalation if needed.

Exceptions / alternate flows

- If raw logs are unavailable, UI clearly states evidence limits and still shows structured events.

UI requirements

- Timeline visualization is preferred over forcing operators to infer sequence from separate tables.
- Offer 'compare sessions' for same SMSC.

6. Live Traffic

- Real-time MT, MO and DLR rates with selectable carrier/SMSC/session filters.
- Current, average, peak and capacity utilization for selected interval.
- Stable live tables: update values in place; do not reorder rows every refresh unless user enables ranking.
- Traffic matrix by carrier/SMSC with queue and delivery quality alongside throughput.
- Historical overlay or baseline/anomaly band when sufficient data exists.

UC-TRF-01 — Investigate unexpected traffic drop

Field	Specification
Primary actor	NOC Operator
Trigger	Low-traffic anomaly alert or visual drop on dashboard.
Preconditions	Historical/current traffic telemetry exists.
Postconditions	Traffic issue is localized or alert is closed as understood.

Main flow

26. User opens Live Traffic at the alert time range.
27. System identifies which carrier/SMSC contributed most to the drop.
28. User checks whether session state, queue, rejection rate or upstream input changed concurrently.
29. User drills into affected connectivity or diagnostics.
30. User acknowledges the alert or performs recovery.

Exceptions / alternate flows

- If the expected baseline is unavailable, label the alert as threshold-based rather than anomaly-based.

UI requirements

- Preserve selected time range when navigating between Traffic, SMSC and Diagnostics.

7. Queue Operations

Queue metric	Requirement
Depth	Current messages waiting
Oldest age	Age of oldest queued message
Ingress/egress	Messages/sec entering and leaving

Growth rate	Net messages/sec and trend
Drain estimate	Calculated from recent stable egress; mark unreliable if rate is volatile/zero
Retries	Messages awaiting retry and retry rate
Expiry	Recent expired count/rate
Destination	Associated carrier/SMSC/route

UC-QUE-01 — Investigate a growing queue

Field	Specification
Primary actor	NOC Operator / Messaging Engineer
Trigger	Queue threshold or growth-rate alert fires.
Preconditions	Queue metrics and destination mapping are available.
Postconditions	Queue is recovering, controlled, or escalated.

Main flow

31. User opens the queue alert.
32. System shows depth, age, growth rate and affected destination.
33. System correlates current egress TPS, session health, throttling and errors.
34. User determines whether cause is carrier outage, capacity restriction, intentional suspension or internal service degradation.
35. User performs allowed action such as reconnect, resume or reroute, or escalates.
36. System tracks whether queue growth stops and estimates drain time.

Exceptions / alternate flows

- If egress is zero, drain time is displayed as unavailable rather than infinity.
- If reroute is unsupported for already queued messages, UI states this explicitly.

UI requirements

- Show queue recovery progress after action.
- Destructive purge, if ever supported, must be separately permissioned and strongly confirmed.

8. DLR & Carrier Quality

- Delivery funnel: submitted → accepted by SMSC → DLR received → delivered/failed/pending.
- Status breakdown: DELIVRD, UNDELIV, EXPIRED, REJECTD, UNKNOWN, pending, no-DLR.
- Latency percentiles: median/P50, P95 and P99 where meaningful.
- Compare carriers/SMSCs over identical time windows.
- Trend quality independently from raw traffic volume.
- Expose carrier-specific DLR/error mappings without forcing users to know vendor codes.

UC-DLR-01 — Investigate deterioration in delivery rate

Field	Specification
Primary actor	NOC Operator / Messaging Engineer
Trigger	Delivery rate falls below threshold or anomaly detector flags change.
Preconditions	DLR and submission data are correlated.
Postconditions	Cause is narrowed to delivery failure, DLR delay/loss, routing/connectivity or unknown.

Main flow

37. User opens DLR Performance from the alert.
38. System compares current delivery rate to prior/baseline period.
39. User identifies affected carrier/SMSC and dominant failure statuses.
40. User checks DLR latency and no-DLR rate to distinguish delayed receipts from true delivery failure.
41. User opens correlated SMPP/carrier errors and events.

42. User optionally traces sample affected messages to validate the pattern.

Exceptions / alternate flows

- If the observation window is too recent for final DLRs, system warns that results are immature.

UI requirements

- Make DLR maturity/window warnings prominent to avoid false incident conclusions.

9. Carrier Routing & Failover

Kamex should expose network-level routes and failover behavior, not customer/commercial routing.

Route detail	Requirement
Match	Destination prefix/network/market and any gateway-level technical conditions
Targets	Primary, secondary and emergency SMSCs
State	Active target and health of alternatives
Capacity	Available/used TPS per target where known
Failover	Automatic/manual mode, trigger, last transition and reason
Traffic	Current route TPS and queue

UC-RTE-01 — Simulate a carrier route

Field	Specification
Primary actor	NOC Operator / Messaging Engineer
Trigger	User needs to know where a destination would be sent without transmitting a message.
Preconditions	Routing data is loaded.
Postconditions	No message is sent; route decision is available for troubleshooting.

Main flow

43. User opens Route Simulator and enters destination/MSISDN plus optional technical attributes.
44. System normalizes the number and identifies market/network/prefix.
45. System evaluates current gateway routing rules.
46. UI shows matched route, selected SMSC, fallback order, health and capacity indicators.
47. System explains why the route matched and whether any target is currently unavailable.

Exceptions / alternate flows

- If no route matches, return a clear 'No route' result and diagnostic reason.

UI requirements

- Simulator must be explicitly labeled non-transmitting.

UC-RTE-02 — Manually fail over a carrier route

Field	Specification
Primary actor	Messaging Engineer / Gateway Administrator
Trigger	Primary SMSC is degraded or carrier instructs traffic movement.
Preconditions	Alternate route is configured and user has failover permission.
Postconditions	Route is operating on selected target with auditable reason.

Main flow

48. User opens route and selects Fail Over.
49. System displays current and proposed target health, capacity, queue implications and affected route.
50. User supplies reason and confirms.
51. System changes active route through the control plane.
52. UI monitors traffic transfer, errors and queue behavior.

53. System records transition and creates an operational event.

Exceptions / alternate flows

- If alternate target is unhealthy or lacks capacity, block or strongly warn according to policy.

UI requirements

- Always show current active path prominently; never hide manual override state.

10. Message Trace

- Search by Kamex message ID, SMSC/carrier message ID, MSISDN and time range subject to privacy permissions.
- Present lifecycle as a chronological trace, not a raw log dump.
- Stages may include ingress, route decision, queue, submit_sm, submit_sm_resp, carrier message ID, deliver_sm/DLR, callback/hand-off where relevant.
- Show per-stage timestamps and calculated latency.
- Link each stage to related structured event/log evidence.
- Mask message content and MSISDN by default; privileged reveal is time-limited and audited.

UC-MSG-01 — Trace a failed or delayed SMS

Field	Specification
Primary actor	Support Engineer / Messaging Engineer
Trigger	A message ID/MSISDN is supplied for investigation.
Preconditions	User has trace permission and sufficient search criteria.
Postconditions	Operator has a defensible explanation of where processing succeeded, failed or remains pending.

Main flow

54. User searches using message identifier or MSISDN/time window.
55. System returns matching messages with masked identifiers and status.
56. User selects a message.
57. System renders end-to-end lifecycle and highlights the first abnormal/missing stage.
58. User inspects route, SMSC response, DLR status and correlated errors.
59. User copies a sanitized diagnostic summary or navigates to carrier/SMSC issue.

Exceptions / alternate flows

- If no message exists, distinguish 'not found' from retention-expired or insufficient permission.
- If multiple messages match an MSISDN, require narrowing rather than guessing.

UI requirements

- Provide copyable IDs and timestamps without exposing credentials or unnecessary PII.

11. SMPP Error Intelligence

- Decode SMPP command status into symbolic name and human explanation.
- Aggregate occurrence count/rate by SMSC/session and time.
- Show first seen, last seen and trend.
- Correlate with TPS, throttling, queue and session events.
- Support carrier/vendor-specific error mappings in addition to SMPP standard codes.
- Provide recommended diagnostic checks, clearly labeled as guidance rather than automated root-cause certainty.

UC-ERR-01 — Investigate carrier throttling

Field	Specification
Primary actor	NOC Operator / Messaging Engineer

Trigger	ESME_RTHROTTLED or equivalent rate rises materially.
Preconditions	Error aggregation and traffic metrics exist.
Postconditions	Throttling is understood and operational response is tracked.

Main flow

60. User opens the throttling alert/error.
61. System shows affected SMSC/session, rate and timeline.
62. System overlays submitted TPS, accepted TPS and queue growth.
63. User compares current throughput to known carrier capacity.
64. User determines whether traffic should be rate-reduced, rerouted or escalated to carrier.
65. User monitors recovery after action.

UI requirements

- Error detail should link directly to affected queue, session and traffic chart.

12. Events, Logs & Correlation

12.1 Structured events

- Connection lost/restored; bind failed/succeeded; session flapping.
- Enquire-link timeout; socket/network failure.
- SMSC suspended/resumed/reconnected.
- Route failover/manual override.
- Queue threshold crossed/recovered.
- DLR degradation/recovery.
- Service restart/failure/recovery.
- Operator actions and control-plane failures.

12.2 Logs

- Live tail with pause, search and severity/component filters.
- Filter by carrier, SMSC, session, message ID, MSISDN (permissioned), service and error code.
- Context window around selected event rather than forcing users to manually locate timestamps.
- Deep-link from message trace/event/SMPP error to filtered logs.
- Clear retention boundaries and indication when older logs are unavailable.

13. Alerts & Incident Experience

Alert category	Examples
Availability	SMSC down; session bind failure; service/node unavailable
Connectivity quality	Session flapping; enquire-link timeout; elevated reconnects
Capacity	High utilization; carrier throttling; queue growth
Delivery quality	DLR rate drop; no-DLR spike; latency increase
Traffic anomaly	Unexpected low/high traffic relative to baseline
Infrastructure	CPU/memory/disk/file descriptors/DB connection pressure
Telemetry	Metrics/log/event stream stale or unavailable

UC-ALT-01 — Acknowledge and work an alert

Field	Specification
Primary actor	NOC Operator
Trigger	An active alert appears.
Preconditions	Alert is active and user has acknowledgement permission.
Postconditions	Alert lifecycle is traceable from detection to recovery.

Main flow

66. User opens alert detail.
67. System shows severity, start time, duration, affected objects, impact metrics and correlated evidence.
68. User acknowledges the alert with optional note.
69. User follows recommended diagnostic links and performs authorized remediation.
70. System automatically updates alert state when recovery conditions are met, or allows policy-based manual resolution.
71. Timeline retains acknowledgement, actions and recovery.

Exceptions / alternate flows

- If the condition recurs after recovery, create/reopen according to incident policy and retain previous history.

UI requirements

- Avoid alert storms by grouping repeated events under one incident where appropriate.
- Show active duration and business/traffic impact, not severity alone.

14. System & Service Monitoring

Object	Key metrics/status
Bearerbox/Kamex core	Process state, uptime, CPU/RAM, message rate, internal health
SMSBox instances	State, uptime, request rate, latency, errors
DLR datastore	Reachability, latency, connections, storage pressure
Database	Reachability, pool usage, slow/error indicators
Metrics collector	Scrape/ingest freshness and failures
Nodes	CPU, memory, disk, network I/O, load, process health, software version

UC-SYS-01 — Diagnose a degraded gateway service

Field	Specification
Primary actor	NOC Operator / Gateway Administrator
Trigger	Service health changes to degraded/critical.
Preconditions	System telemetry is available.
Postconditions	Service is recovered or incident is escalated with evidence.

Main flow

72. User opens System from alert/dashboard.
73. System identifies affected service/node and dependency status.
74. User checks resource pressure, recent restart/crash events and dependent services.
75. User opens related logs or performs an authorized restart if policy permits.
76. UI monitors service recovery and downstream SMSC/traffic impact.

Exceptions / alternate flows

- If the monitoring subsystem itself is affected, label dependent statuses Unknown.

UI requirements

- Dependency visualization should distinguish root service failure from downstream symptoms.

15. Diagnostic Test Tools

Tool	Expected behavior
SMPP connectivity test	Validate TCP reachability and, where safe/supported, controlled bind diagnostics without exposing password
Test SMS	Send an explicitly tagged operational test through selected

	route/SMSC with audit record
Route simulator	Resolve destination and route without transmission
DLR lookup	Search carrier message ID/Kamex ID and show receipt lifecycle
Encoding/segment analyzer	Detect GSM-7/UCS-2, character count, segment count and payload implications
Number/prefix lookup	Normalize number and identify configured market/network/prefix

UC-TST-01 — Send a controlled test SMS

Field	Specification
Primary actor	Messaging Engineer
Trigger	Engineer needs to verify live end-to-end delivery.
Preconditions	User has test-send permission; destination policy permits test.
Postconditions	Test result is traceable and audited.

Main flow

77. User opens Test SMS and selects SMSC/route or allows route resolution.
78. User enters approved test destination and message.
79. System shows expected encoding, segment count and selected route.
80. User confirms transmission.
81. System sends the test and immediately opens its message trace.
82. UI follows submit response and DLR until final status or timeout.

Exceptions / alternate flows

- If test sending is disabled in the environment, UI explains policy and leaves non-transmitting diagnostics available.

UI requirements

- Visually distinguish test traffic from production traffic in traces/events.

16. Audit & Operational Governance

- Audit all reconnect, suspend/resume, failover, service restart, test-send, alert acknowledgement and privileged PII reveal actions.
- Record actor, role, timestamp, target object, previous state, requested action, reason, result and correlation/incident ID where available.
- Audit data is read-only in the UI and filterable by actor, action, object and time.
- High-risk actions should support maker-checker/approval later without redesigning the interaction model.

17. UX State Model & Interaction Rules

State	UI treatment
Loading	Skeletons for cards/tables; do not show zero values that look real
Live	Freshness timestamp/stream indicator
Stale	Warning banner and age of last successful telemetry
Empty	Explain why no data exists and next useful action
Partial	Show available data plus explicit unavailable components
Error	Actionable error message with retry/correlation ID where useful
In progress	Disable duplicate destructive action and stream progress
Permission denied	Explain required permission without revealing sensitive data

17.1 Tables

- Server-side filtering/pagination for high-volume event/log/message datasets.
- Pin operationally important columns; allow column chooser for engineering detail.

- Persist user filters per screen where appropriate.
- Do not encode health by color alone; include text/icon/state label.
- Allow CSV export only where policy permits and record export of sensitive operational data.

17.2 Time and refresh

- Use absolute timestamp plus relative age on hover/detail.
- Default operational dashboards to short windows; analytical views retain selectable history.
- Pause live refresh automatically while user is selecting/copying table content where necessary.
- Display metric sampling interval and telemetry freshness in diagnostics.

18. Non-Functional Requirements for the UI

Area	Requirement
Performance	Dashboard initial meaningful render target ≤2s on normal internal network; interactions should acknowledge within ~200ms where local/UI-only.
Real-time	Critical state changes should surface within telemetry/control-plane limits; UI must show freshness.
Availability	Read-only monitoring should degrade gracefully if control actions are unavailable.
Security	RBAC; no credentials in browser payloads; CSRF/session protections; privileged action re-authentication where policy requires.
Privacy	Mask MSISDN/content; configurable retention; audited reveal/export.
Accessibility	Keyboard navigation, semantic labels, non-color status cues, sufficient contrast.
Responsive	Primary target desktop ≥1280px; usable on tablet for monitoring; wallboard mode for dashboard.
Observability	Frontend/API failures carry correlation IDs and are themselves monitorable.

19. Screen-Level Acceptance Checklist

- Can the operator identify current health and data freshness immediately?
- Can every degraded state be drilled into without losing time/filter context?
- Does the screen distinguish symptom, impact and likely evidence?
- Are all operational actions state-aware, permission-aware and auditable?
- Are protocol codes translated into human-readable meaning while retaining raw values?
- Can the operator reach related events/logs/message traces in one or two clicks?
- Are unknown/stale states represented honestly?
- Is sensitive information masked by default?
- Can a user recover from an action failure without refreshing the entire application?
- Does the design avoid exposing Kannel configuration-file concepts unless explicitly in engineering diagnostics?

20. Suggested Redesign Delivery Sequence

Phase	Screens	Why
1 — Operational foundation	Global shell; Dashboard; Alerts; Carrier/SMSC list/detail; SMPP Sessions	Defines design system and core operational hierarchy
2 — Traffic assurance	Live Traffic; Queues; DLR Performance	Adds capacity/congestion and quality workflows
3 — Troubleshooting	Message Trace; SMPP Errors; Events; Logs	Makes the console materially useful for support/NOC diagnosis
4 — Network control	Carrier Routes; Failover; Route Simulator; controlled SMSC actions	Adds safe operational intervention
5 — Platform operations	Services; Nodes; Performance; Audit; Test	Completes gateway operations experience

	Tools	
6 — Intelligence	Anomaly detection; cross-signal correlation; incident summaries; capacity forecasting	Build after trustworthy telemetry and workflows exist

21. Core Domain Model for UI/API Alignment

Entity	Key relationships
Carrier	Has many SMSCs; aggregates quality/traffic/incidents
SMSC	Belongs to carrier; has many sessions; owns/receives queues; route target
SMPP Session	Belongs to SMSC; emits protocol counters/events/errors
Carrier Route	Matches network/prefix; has ordered SMSC targets; tracks active target
Queue	Associated with destination/SMSC/route; exposes depth/age/rates
Message Trace	References route, SMSC, carrier message ID, DLR and related events
DLR	References message and carrier/SMSC; has raw and normalized status
Operational Event	References one or more domain objects and correlation ID
Alert/Incident	Groups condition/events, affected objects, acknowledgement and recovery
Service/Node	Hosts gateway components and produces health/performance telemetry
Audit Entry	Records privileged user action and result

22. Definition of Done for UI Redesign

The redesign is complete when a NOC operator can monitor the gateway, identify an unhealthy carrier/SMSC/session, understand traffic and queue impact, diagnose common SMPP/DLR faults, safely execute permitted recovery actions, trace an individual message, and reconstruct the incident from events/audit without editing Kannel configuration files or routinely using SSH/log-tail commands.